

Prototype Shopping List

Steel Beam Cutting Cell - 1:5 desktop prototype | Ravi Mahadeva | 04 October 2026

Stage	About
Stage 1 - Frame and bed	£340
Stage 2 - Motion and controller	£380
Stage 3 - Arms and tools	£540
Stage 4 - Cameras (vision)	£100
Stage 5 - Safety and enclosure	£410
Stage 6 - Electrical	£110
Consumables	£40
Total	about £1,920

Before you buy

- Prices are approximate (GBP, October 2026) and change - check each one before ordering.
- You already have the **Jetson Orin Nano**; it is not in the total.
- Order the aluminium **cut to size** (lengths on the last page) - it saves a lot of work.
- Buy and wire the **safety parts (stage 5) before any motor moves**: the E-stop and safety relay must cut motor power. Have the mains wiring checked by a competent person.
- Cheapest way to start: stages 1 and 2 first (frame and moving gantry, about £720), then arms, cameras and the rest.
- Save on the arms by buying the STS3215 servos and printing the SO-101 parts yourself (about £130 per arm instead of £240). A used Pilz PNOZ s3 is about £75.
- No real cutting on the prototype: the Cutter holds a **pen** that marks the cut lines.

Stage 1 - Frame and bed

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	V-slot 20x40 aluminium extrusion	cut to size: 4 x 1500, 6 x 600, 4 x 580 mm (cad/prototype_cut_list.csv) lengths come from the CAD model	11.9 m	£14	£167	Ooznest (cut to size), Amazon UK
	V-slot 20x20 aluminium extrusion	cut to size: 2 x 1300, 2 x 300, 6 x 100, 1 x 720 mm bed rails and legs; Z axes; camera post	4.5 m	£8	£36	Ooznest
	V-slot 20x60 aluminium extrusion	cut to size: 2 x 660 mm the two bridges	2	£16	£32	Ooznest
	Corner brackets + M5 T-nuts + M5 bolts	40 cast corner brackets, 200 drop-in T-nuts, M5x8/M5x10 button heads	1 kit	£45	£45	Ooznest, Amazon UK
	Rubber feet / levelling feet	M8 levelling feet for the base rails stops it walking on the bench	4	£3	£12	Amazon UK
	Bed rollers	20 mm aluminium tube x 100 mm on 8 mm silver steel shaft, 2 x 608 bearings each or 3D print the rollers	6	£4	£24	Amazon UK, Simply Bearings
	Roller brackets	3D printed in PETG (12 off)	12	£0.50	£6	print
	Outfeed table board	6 mm plywood or MDF 1300 x 180 mm	1	£8	£8	builders' merchant
	Scrap tray	folded aluminium sheet or 3D printed tray 1300 x 90 mm	1	£10	£10	print / sheet
Stage total					£340	

Stage 2 - Motion and controller

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	V-wheel gantry plate kits	20-series gantry plate + 4 V-wheels (625 bearings) + eccentric spacers 2 per bridge end x 2 bridges + 1 per carriage x 2	6	£18	£108	Ooznest
	NEMA 17 stepper motor	1.5-2 A, 45-48 mm body, 5 mm shaft X (one per bridge) and Y (one per carriage)	4	£12	£48	Ooznest, StepperOnline
	NEMA 17 with integrated T8 lead screw	T8 x 2 mm lead, 300 mm screw, with brass anti-backlash nut Z axes	2	£25	£50	StepperOnline, Amazon UK
	GT2 belt 6 mm	10 m reel (8 m needed: 2 per bridge along the rails, 1 per carriage)	1	£12	£12	Ooznest
	GT2 20T pulleys + idlers	4 x 20T 8 mm bore (cross shafts), 2 x 20T 5 mm bore (Y), 8 idlers	1 set	£25	£25	Ooznest
	8 mm cross shafts + pillow blocks	2 x 700 mm 8 mm silver steel, 4 x KFL08 bearings, 2 x 5-8 mm couplers one motor drives both sides of each bridge	1 set	£25	£25	Amazon UK, Simply Bearings
	Home / limit switches	mechanical microswitch with lever, NC wiring one per axis + 2 spare	8	£1.50	£12	Amazon UK
	FluidNC 6-axis CNC controller (ESP32)	6 driver sockets, USB to the Jetson (e.g. Elecrow / Bart Dring 6x CNC Controller) flash once from the web installer - no programming	1	£40	£40	Elecrow, Tindie
	TMC2209 stepper driver modules	StepStick format, UART or standalone quiet; one per axis	6	£5	£30	Amazon UK, Ooznest
	Shielded 4-core motor cable + drag chain	10 m 4 x 0.5 mm ² shielded, 2 m 10x15 drag chain	1 set	£30	£30	Amazon UK, RS
Stage total					£380	

Stage 3 - Arms and tools

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	SO-101 6-servo arm kit (follower)	6 x Feetech STS3215 serial-bus servos + bus servo driver board; print the arm parts in PLA ~£120-150 each if you buy the servos and print everything yourself	2	£240	£480	Unmanned Tech UK (servo kit Pro, unprinted), Seeed
	Bus servo adapter	Waveshare Bus Servo Adapter (A) - USB to the Jetson, one per arm often included in the kit	2	£5	£10	Waveshare, The Pi Hut
	12 V power supply for the servos	Mean Well LRS-100-12 (12 V 8.5 A) - check your servos are the 12 V version	1	£16	£16	RS, Farnell
	Cutter 'torch' for the prototype	pen in a sprung 3D-printed holder (marks every cut line on the model beam) no real cutting on the prototype: see docs/PROTOTYPE.md	1	£3	£3	print
	Handler magnet	12 V 25 mm lifting electromagnet (~2.5 kg hold) + MOSFET module + flyback diode switch it from a FluidNC output	1	£12	£12	Amazon UK
	Model beams	3D printed UB 305x165 at 1:5 (61 x 33 mm), 1 m long, with 0.5 mm steel strip glued on the top flange the magnet needs steel to grip	4	£4	£16	print
Stage total					£537	

Stage 4 - Cameras (vision)

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	USB wide-angle camera (safety zone)	1080p UVC, 90-120 deg view (e.g. Logitech C920/C922 or an Arducam USB wide-angle) plug and play on the Jetson; YOLO watches the cell	1	£50	£50	Amazon UK, Pimoroni
	CSI camera (close-up of the Cutter)	Arducam IMX219 for Jetson Orin Nano (22-pin cable), wide-angle lens optional; checks the marked lines	1	£25	£25	Arducam, RobotShop UK
	Powered USB 3 hub	4-port, own power supply controller + 2 servo adapters + camera	1	£20	£20	Amazon UK
Stage total					£95	

Stage 5 - Safety and enclosure

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	Emergency stop station	yellow box, red 40 mm mushroom, turn to release, 2 NC contacts (e.g. Schneider Harmony XALK178 family) BS EN ISO 13850; one NC to the safety relay channel 1 (and 2) - add an NO/NC block for the Jetson	1	£42	£42	RS, Farnell, TME
	Safety relay	Pilz PNOZ s3 24 V DC (dual channel, manual reset) or equivalent cuts the 24 V motor and 12 V servo supplies - this is how a real machine does it	1	£180	£180	RS, Farnell (new) / ~£75 used
	Door interlock switch	non-contact coded magnetic safety switch, NC (e.g. Pilz PSEN or Schmersal BNS) BS EN ISO 14119; wired into the safety relay	1	£60	£60	RS, Farnell
	Reset push button	22 mm blue illuminated, 1 NO outside the enclosure with a view inside	1	£8	£8	RS, Amazon UK
	Stack light	24 V LED tower, red / amber / green / blue driven by the Jetson through a relay module	1	£25	£25	Amazon UK
	4-channel relay module (opto-isolated)	3.3 V logic input, 24 V contacts stack light and magnet	1	£8	£8	Amazon UK
	IR break-beam sensor pair	5 mm IR break-beam (demonstrates the light-curtain input) NOT a safety light curtain - demo only	1	£6	£6	Pimoroni, The Pi Hut
	Polycarbonate sheet 3 mm	front, door and ends (~1.5 m2) - polycarbonate, not acrylic (acrylic shatters)	1.5 m2	£40	£60	plastics stockist
	Hinges + handle + 20x20 door frame	2 hinges, 1 handle, 2.4 m 20x20	1 set	£25	£25	Ooznest
Stage total					£414	

Stage 6 - Electrical

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	24 V power supply	Mean Well LRS-150-24 (24 V 6.5 A) - motors, controller, safety circuit	1	£16	£16	RS (0424827)
	IEC mains inlet with switch and fuse	panel mount, 5 A fuse have the mains side checked by a competent person	1	£8	£8	RS, Farnell
	DIN rail + terminals + fuse holders	35 mm DIN rail, 20 terminals, 4 x 5x20 fuse holders, end stops	1 set	£30	£30	RS, Farnell
	Control box	ABS or steel enclosure ~300 x 200 x 150 with cable glands holds the Jetson too	1	£30	£30	RS, CPC
	Wire + ferrules + crimp tool	0.75 and 1.5 mm ² flexible, ferrule kit	1 set	£30	£30	Amazon UK, CPC
	Jetson Orin Nano Super (4 GB)	you already have it - docs/JETSON_SPECS.md runs the app, YOLO and the planner	1	£0	£0	-
Stage total					£114	

Consumables

OK	Item	What to look for	Qty	Each	Line total	Where (UK)
	PETG and PLA filament	1 kg each brackets, arm parts, model beams	2	£20	£40	Amazon UK, 3DJake
Stage total					£40	

Aluminium cut list (order cut to size)

OK	Profile	Length (mm)	Quantity	Used for
	20x20 V-slot	100	6	bed leg
	20x20 V-slot	300	2	Z axis
	20x20 V-slot	720	1	camera post
	20x20 V-slot	1300	2	bed rail
	20x40 V-slot	580	4	cross member
	20x40 V-slot	600	6	upright
	20x40 V-slot	1500	2	base rail
	20x40 V-slot	1500	2	top rail
	20x60 V-slot	660	2	bridge

The lengths come from the CAD model (cad/prototype_1to5.step). The build steps, the safety wiring and why each part was chosen are in docs/PROTOTYPE.md.